

Bike Sharing Demand Forecasting - Exploratory Data Analysis Report

This report presents an Exploratory Data Analysis (EDA) and forecasting overview for the Bike Sharing Dataset from the UCI Machine Learning Repository. The objective is to predict hourly bike rental demand (`cnt`) using weather, seasonal, and temporal features to support operational planning and bicycle logistics.

1. Business Problem

A bicycle rental company requires accurate hourly demand forecasts to optimize bicycle allocation, staffing, and operational logistics. Forecasting demand helps reduce shortages, improve service quality, and support data-driven planning.

2. Dataset Overview

The dataset contains hourly bike rental records along with weather conditions, seasonal indicators, working day flags, and environmental variables such as temperature, humidity, and windspeed.

3. Exploratory Data Analysis

Exploratory analysis revealed strong temporal and seasonal demand patterns. Demand peaks during commuting hours and working days. Summer and Fall exhibit higher bike usage, while poor weather conditions significantly reduce rentals.

4. Feature Engineering

Important engineered features include peak-hour indicators, weekend flags, cyclical time features (sine/cosine transformations), and weather severity indicators. These features improve forecasting accuracy by capturing nonlinear patterns.

5. Forecasting Models

Three forecasting models were evaluated: - Linear Regression - Random Forest Regressor - XGBoost Regressor. XGBoost was selected as the final production model due to its superior ability to model nonlinear relationships, seasonality, and weather-driven demand behavior.

6. Model Performance

The selected XGBoost model achieved strong predictive performance with low forecasting error and high explanatory power. Key Metrics: - Mean Absolute Error (MAE): ~21 - Root Mean Squared Error (RMSE): ~31 - R^2 Score: ~0.95

7. Business Insights

Key insights include: - Weather significantly impacts bike demand. - Demand spikes occur during commuting hours. - Seasonal effects strongly influence rentals. - Short-term forecasting (24–72 hours) provides the highest operational reliability.

8. Operational Recommendations

Recommended operational practices include: - Refresh forecasts every 1–3 hours. - Retrain models seasonally. - Monitor weather-driven demand fluctuations. - Use automated model monitoring in production.

9. Production Considerations

The project follows production-ready machine learning engineering practices including modular architecture, logging, automated testing, reusable utilities, exception handling, and scalable pipeline design.

10. Conclusion

The Bike Sharing Demand Forecasting project successfully demonstrates how machine learning can improve operational planning and bicycle logistics. The XGBoost forecasting model provides reliable short-term demand predictions and strong business value for real-world deployment.

Project: Bike_Sharing_Demand_Forecasting

Prepared Using: Python, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn

Purpose: Exploratory Data Analysis & Operational Forecasting